

EYEABETICS

RESEARCH, TECHNICAL SCOPE & OCULOMICS ROADMAP

SIH 2026 | Biomedical AI Research Report

SCREEN

EXPLAIN

CONNECT

RESEARCH POSITIONING

Eyeabetics is a research prototype for AI assisted diabetic retinopathy screening, retinal structural evidence, referral support, and longitudinal continuity. Oculomics is positioned as a future research layer, not as a current capability for diagnosing systemic diseases.

1. Executive Summary

Eyeabetics is a retinal AI screening prototype developed to support screening in resource-constrained healthcare settings. The system brings together image quality assessment, DR classification, lesion and vessel analysis, referral support, patient records, and a web interface in one workflow. The system supports screening and referral workflows and does not replace professional ophthalmic diagnosis.

The International Diabetes Federation estimated that approximately 77 million adults in India had diabetes in 2019. The Indian National Survey of Blindness and Visual Impairment, conducted from 2015 to 2019, reported diabetic retinopathy in 16.9% of assessed people with diabetes and sight threatening DR in 3.6%. These figures come from different sources and periods and should therefore be interpreted separately.

2. Problem Context

Diabetic retinopathy can progress before patients notice visual symptoms. A practical screening system needs to do more than classify an image. It must address image quality and gradeability, retinal evidence, referral, continuity of records, follow up, and practical constraints such as limited specialist availability and intermittent connectivity.

THE CENTRAL PROBLEM

How can retinal screening move from an isolated image and a single decision toward an evidence supported, referral aware, and longitudinal workflow for underserved populations?

3. Eyeabetics Current Solution

ID	CAPABILITY	WHAT IT DOES
01	Image Quality Assessment	Checks whether the captured fundus image is sufficiently gradeable. The current prototype does not claim adaptive image enhancement.
02	DR Classification	EfficientNet-B0 based five-level DR classification, ranging from Grade 0 to Grade 4.
03	Lesion Segmentation	U-Net based pixel-level structural analysis of retinal lesion patterns. The model identifies different lesion classes at the pixel level.
04	Vessel Segmentation	U-Net based retinal vessel segmentation that provides supporting structural measurements, such as vessel area.
05	Risk and Referral	Combines the DR grade with selected patient factors to generate a prototype referral category.
06	Retina Passport	Longitudinal patient records and screening history intended to support follow up rather than isolated screening.
07	Low-Connectivity Workflow	Offline referral and queue concepts designed for environments where connectivity may be intermittent.

4. Technical Architecture

**FUNDUS IMAGE → QUALITY ASSESSMENT → DR CLASSIFIER → LESION U-NET → VESSEL U-NET
→ STRUCTURAL ANALYSIS → RISK / REFERRAL → REPORT + PATIENT RECORD**

TECHNOLOGY STACK

Python and PyTorch for model development; ONNX for model interchange; MATLAB R2026a for inference; Node.js for the backend; MongoDB for persistence; React and Vite for the interface; and Simulink for planned workflow and scalability simulation.

5. Evidence and Validation Boundaries

- **Eyeabetics is currently a research prototype** demonstrating an integrated retinal screening workflow. The results show technical feasibility, not clinical validation. Do not claim target sensitivity, specificity, calibration, prospective validation, expert validation, Grad-CAM validation or annual throughput unless those measurements have actually been completed.
- **Clinical validation is still required**, including testing across different populations, devices, hospitals, and patient groups before real-world deployment
- **Image quality assessment checks whether a fundus image is suitable for analysis.** It does not enhance or improve low-quality images.

- **The structural outputs provide supporting retinal evidence** and should not be treated as a direct explanation of the AI model's decision.
- **Future work** will focus on external validation, calibration, prospective testing, privacy, security, and regulatory requirements.

6. Oculomics: What It Means

Oculomics is an emerging research area that studies whether information from the eye and retina can provide useful signals about health beyond eye disease. Recent research and reviews describe work spanning cardiovascular, cerebrovascular, renal, metabolic and neurodegenerative areas.

The retina is particularly useful for this research because it provides a non-invasive view of small blood vessels and neural structures. However, an association between a retinal feature and a systemic condition does not by itself establish a diagnosis. Population diversity, camera variation, image quality, confounding, calibration and external validation remain central research challenges.

EYEABETICS OCULOMICS PRINCIPLE

Retina → measurable biomarkers → validated associations → risk stratification research → clinician review. Not retina → autonomous systemic diagnosis.

7. Retinal Biomarkers Relevant to Oculomics

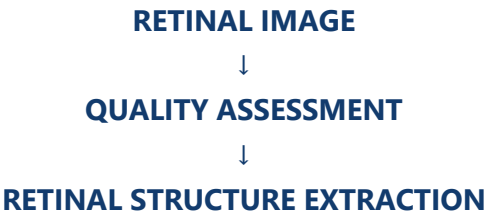
BIOMARKER	RESEARCH RELEVANCE
Vessel caliber	Can provide information about retinal vascular changes
Vessel density	Measures the distribution of vessels in the retina
Vessel geometry	Helps describe retinal vessel structure
Lesion patterns	Provides structural evidence related to diabetic retinopathy
Longitudinal change	Allows retinal features to be studied over time

8. Future Oculomics Direction

The current Eyeabetics system focuses on diabetic retinopathy screening and retinal structural analysis. In future work, these measurable retinal features could be studied alongside clinical data to investigate possible associations with broader systemic health.

This would be a research extension rather than a current diagnostic capability. Any such application would require appropriate datasets, clinical validation, external testing, and expert review.

9. Proposed Future Research Architecture





10. Research Roadmap

PHASE	FOCUS	OUTPUT
Phase 1	Standardize retinal phenotypes	Define reproducible vessel, lesion and structural measurements.
Phase 2	Build labeled research cohorts	Link retinal imaging with carefully defined systemic endpoints and covariates.
Phase 3	Develop risk models	Compare models based on the retina and multimodal models with appropriate baselines.
Phase 4	External validation	Test across devices, sites and relevant population strata.

11. Key India Evidence

EVIDENCE	VALUE	INTERPRETATION
Adults with diabetes in India, IDF 2019	~77 million	National estimate for adults with diabetes.
Any DR among assessed people with diabetes, National Survey 2015 to 2019	16.9%	Survey estimate among assessed people with diabetes.
Sight threatening DR, same survey	3.6%	Survey estimate; not equivalent to total DR prevalence.
Rural vs urban DR in the same survey	15.5% vs 20.7%	Contextual comparison within that survey population.

12. Ethics, Privacy and Deployment

- Retinal images and patient metadata should be handled as sensitive health information with appropriate access control and retention policies.
- Consent, data minimization, secure transmission, auditability and role based access should be designed before clinical deployment.
- AI outputs should be communicated as screening or decision support information, with uncertainty and clear referral guidance.
- Systemic risk research must avoid presenting correlations as diagnoses and should include appropriate clinical oversight.
- External validation should represent the intended deployment population, devices and workflow.

13. Current vs Future Capability

CAPABILITY	CURRENT PROTOTYPE	FUTURE RESEARCH
DR screening	Implemented prototype workflow	External and prospective clinical validation
Image quality	Quality assessment and gradeability	Robust multi device quality validation
Lesion evidence	U-Net structural evidence	Validated lesion biomarkers and longitudinal studies
Vessel evidence	U-Net vessel segmentation	Quantitative vascular biomarker research
Referral	Prototype risk and referral logic	Clinical threshold validation and workflow studies
Retina Passport	Patient history and progression concept	Longitudinal outcomes research
Oculomics	Not a current diagnostic feature	Systemic risk research using validated retinal biomarkers
Simulink	Planned workflow and scalability layer	Measured capacity, latency and network simulations

14. Selected Research and Reference Links

- **International Diabetes Federation, Diabetes Atlas**
<https://diabetesatlas.org/resources/previous-editions/>
- **WHO India, Diabetes**
<https://www.who.int/india/health-topics/diabetes>
- **National Survey of Blindness and Visual Impairment, India, 2015 to 2019**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8725073/>
- **AIOS and VRSI diabetic retinopathy screening guidance**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7942107/>
- **AI and retinal imaging for diabetes complications, systematic review**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11883405/>
- **AI-enhanced retinal imaging and systemic disease biomarkers**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11905132/>
- **AI oculomics review, 2025**
<https://pubmed.ncbi.nlm.nih.gov/40910901/>
- **Oculomics-based AI systematic review**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12789765/>
- **Retinal imaging biomarkers for cardiovascular disease, Nature Reviews Cardiology**
<https://www.nature.com/articles/s41569-024-01060-8>
- **IDRiD dataset**
<https://idrid.grand-challenge.org/Data/>
- **IDRiD dataset paper**
<https://www.mdpi.com/2306-5729/3/3/25>

15. Eyeabetics Research Positioning Statement

CURRENT

AI assisted DR screening, retinal structural evidence, risk and referral support, and longitudinal continuity.

FUTURE

A validated Oculomics research layer using measurable retinal biomarkers and multimodal data to investigate

systemic risk associations.

CLINICAL BOUNDARY

Research and screening support. Not autonomous diagnosis of systemic disease and not a replacement for professional medical assessment.